HRMA

Rocket Motor Analysis & Design Suite

Hybrid • Solid • Liquid Rocket Motors

User Guide

A step-by-step worked example

Version 2.6.25 | July 2026

Developed by: Berke Tezgocen • Idea & Testing: Ayberk Cem Aksoy

Scope. HRMA is a preliminary-design and educational tool built on closed-form and one-dimensional engineering correlations. It is *not* a flight-qualification tool. Predicted performance should be cross-checked against an independent code (CEA / RPA / openMotor) and verified by physical testing before firing any motor. See "Scope and Limitations" at the end of this guide.

Contents

1. What is HRMA?

HRMA (Rocket Motor Analysis Suite) is a desktop application for the preliminary design and analysis of hybrid, solid, and liquid rocket motors. It runs a local calculation engine (Flask) inside its own native window and presents results in a dark-themed, interactive interface: an interactive 3D model built live from the solver output, an Analysis Deck of engineering panels, and working CAD / drawing / report exports.

Who is it for? Rocket teams, undergraduate and graduate students, faculty, and anyone who wants to see motor physics end to end. The program does not design a motor *from scratch* for you; it produces geometry, performance and warnings from the parameters you enter, which you then judge with engineering reasoning.

> **Everything runs locally and offline.** None of your design data leaves your computer. The only time the program reaches the internet is at startup, when it asks GitHub whether a newer version exists; if that fails, the program continues to work normally.

1.1 How to use this guide

The guide follows a single **example motor**: a hybrid using HTPB fuel and liquid N_2O oxidizer, 1000 N thrust, 10 s burn time. The same steps apply to solid and liquid motors; the differences are noted in their sections. All screenshots are taken directly from the application.

2. Installation and First Launch

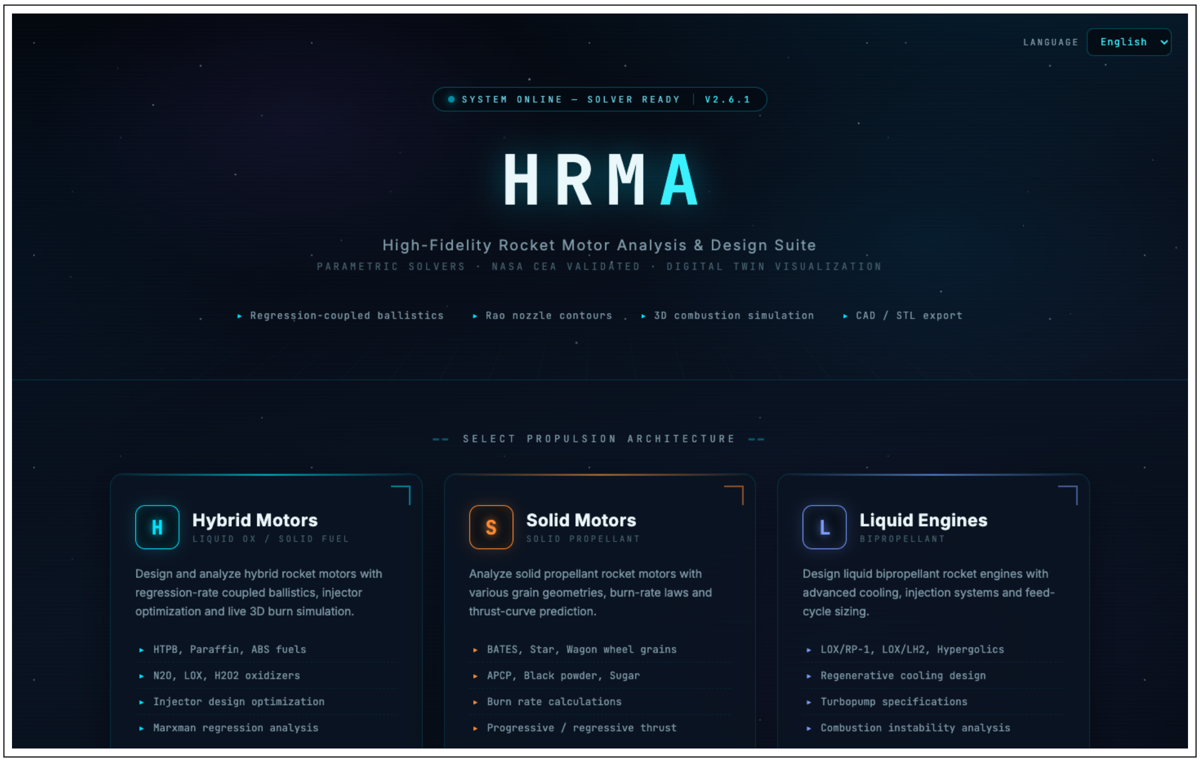
2.1 Installation

- **Windows 10/11:** Download `HRMA-Setup-2.6.25.exe`, double-click and follow the wizard (Next, Next, Install). The installer is per-user; no administrator rights are required. Windows SmartScreen may warn because the installer is unsigned: click "More info", then "Run anyway".
- **macOS 11+ (Apple Silicon):** Open `HRMA-Setup-2.6.25-macOS.dmg` and drag `HRMA` into `Applications`. On first launch, right-click the app and choose "Open" (Gatekeeper requires this once for unsigned apps).

Python and all libraries are bundled; no separate installation is needed.

2.2 First launch

The app shows a splash screen within about a second while the calculation engines load in the background, then opens the home page in its own window. The home page has three motor-type cards and a link to the formula reference.



The home page: cards to choose a motor type (Hybrid, Solid, Liquid), with the language selector, Release Notes, User Guide and Settings links in the top bar.

Language. The selector in the top-right switches the interface between English and Turkish; the choice applies to every page, including chart text. A Turkish edition of this guide ships with the application too.

3. Main Pages

The application has five main pages:

Page	URL	Purpose
Home	`/`	Motor-type selection and formula reference
Hybrid Designer	`/hybrid`	Full hybrid motor design (the most complete page)
Solid Designer	`/solid`	Grain geometry, ballistics, Monte Carlo
Liquid Designer	`/liquid`	Feed system, injector, regen cooling, cycle
Formulas	`/formulas`	The equations used by the solvers (MathJax)

All three designer pages share the same structure: an input form, a Calculate button, a results HUD, standalone panels (transient / injector / 6-DOF), the Analysis Deck, the 3D model, and export buttons.

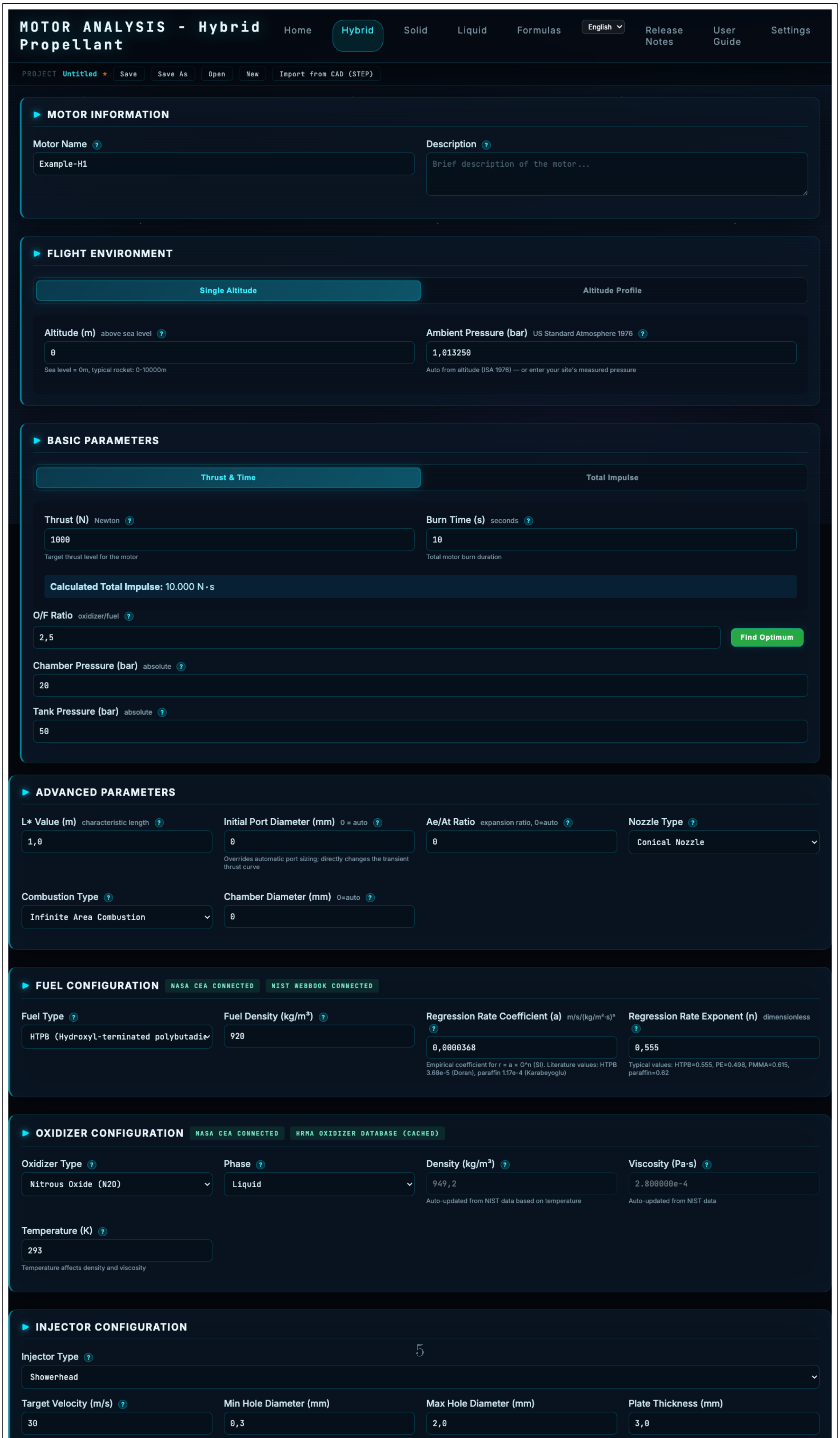
4. Worked Example: Designing a Hybrid Motor

In this section we work through a complete example. Open the `/hybrid` page.

4.1 Step 1 — Enter the parameters

The form comes with sensible defaults, top to bottom. For our example we use these values (most are already the defaults):

Field	**Value**	**Note**
Motor Name	`Example-H1`	Used in reports and exports
Altitude	`0 m`	Sea level (ambient pressure auto)
Thrust	`1000 N`	Target thrust level
Burn Time	`10 s`	Total impulse is derived from this
O/F Ratio	`2.5`	Oxidizer/fuel mass ratio
Chamber Pressure	`20 bar`	Combustion chamber pressure
Tank Pressure	`50 bar`	Must exceed chamber pressure enough
Fuel	`HTPB`	Regression coefficients auto-filled
Oxidizer	N_2O (liquid)	Density updates with temperature
Injector	`Showerhead`	Type-specific parameters

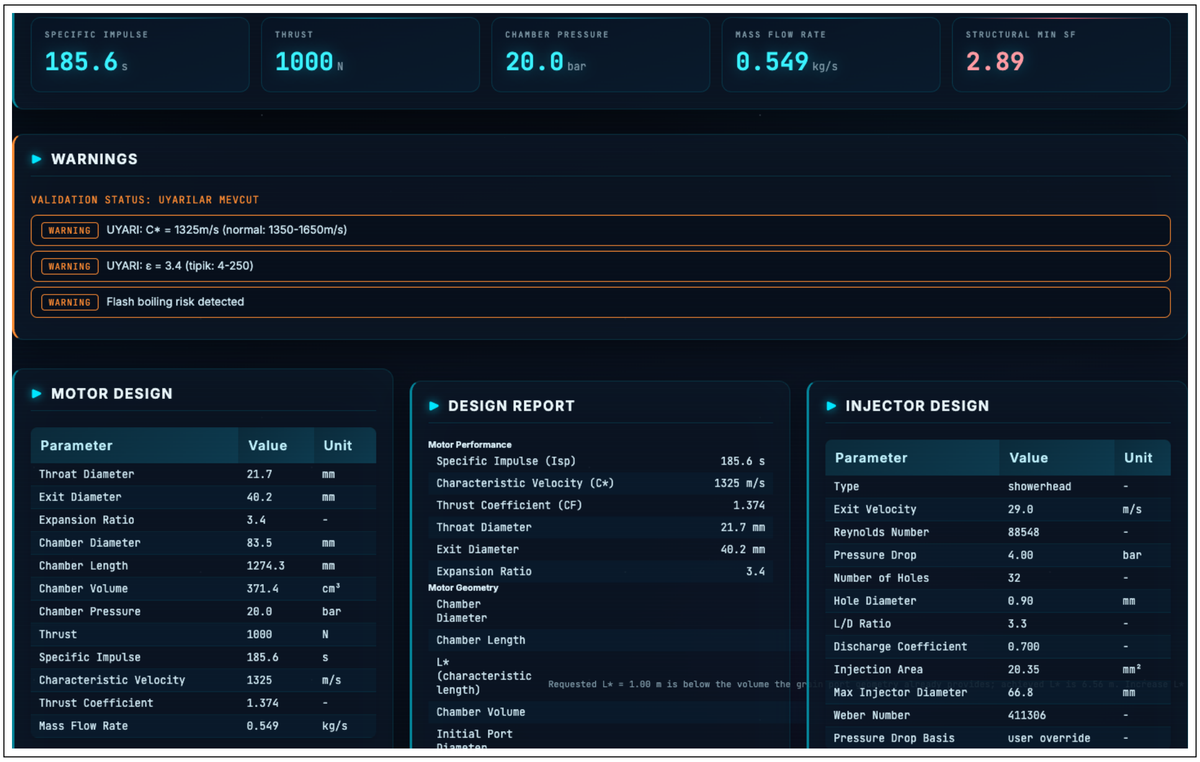

MOTOR ANALYSIS - Hybrid Propellant
Home
Hybrid
Solid
Liquid
Formulas
English
Release Notes
User Guide
Settings
PROJECT Untitled *
Save
Save As
Open
New
Import from CAD (STEP)
MOTOR INFORMATION
Motor Name
Example-H1
Description
Brief description of the motor...
FLIGHT ENVIRONMENT
Single Altitude
Altitude Profile
Altitude (m) above sea level
0
Sea level = 0m, typical rocket: 0-10000m
Ambient Pressure (bar) US Standard Atmosphere 1976
1,013250
Auto from altitude (ISA 1976) — or enter your site's measured pressure
BASIC PARAMETERS
Thrust & Time
Total Impulse
Thrust (N) Newton
1000
Target thrust level for the motor
Burn Time (s) seconds
10
Total motor burn duration
Calculated Total Impulse: 10.000 N·s
O/F Ratio oxidizer/fuel
2,5
Find Optimum
Chamber Pressure (bar) absolute
20
Tank Pressure (bar) absolute
50
ADVANCED PARAMETERS
L* Value (m) characteristic length
1,0
Initial Port Diameter (mm) 0 = auto
0
Overrides automatic port sizing; directly changes the transient thrust curve
Ae/At Ratio expansion ratio, 0=auto
0
Nozzle Type
Conical Nozzle
Combustion Type
Infinite Area Combustion
Chamber Diameter (mm) 0=auto
0
FUEL CONFIGURATION
NASA CEA CONNECTED
NIST WEBBOOK CONNECTED
Fuel Type
HTPB (Hydroxyl-terminated polybutadie
Fuel Density (kg/m³)
920
Regression Rate Coefficient (a) m/s/(kg/m²·s)ⁿ
0,0000368
Empirical coefficient for r = a × G^n (SI). Literature values: HTPB 3.68e-5 (Doran), paraffin 1.17e-4 (Karabeyoglu)
Regression Rate Exponent (n) dimensionless
0,555
Typical values: HTPB=0.555, PE=0.498, PMMA=0.615, paraffin=0.62
OXIDIZER CONFIGURATION
NASA CEA CONNECTED
HRMA OXIDIZER DATABASE (CACHED)
Oxidizer Type
Nitrous Oxide (N2O)
Phase
Liquid
Density (kg/m³)
949,2
Auto-updated from NIST data based on temperature
Viscosity (Pa·s)
2.800000e-4
Auto-updated from NIST data
Temperature (K)
293
Temperature affects density and viscosity
INJECTOR CONFIGURATION
Injector Type
Showerhead
Target Velocity (m/s)
30
Min Hole Diameter (mm)
0,3
Max Hole Diameter (mm)
2,0
Plate Thickness (mm)
3,0

The input form. The "Find Optimum" button sweeps the O/F ratio for the c-optimal value. The "?" next to each field shows a short explanation.*

0 = automatic. Enter `0` into fields such as expansion ratio, chamber diameter or port diameter and the solver sizes them itself. Override the defaults with your own data (for example regression coefficients fitted from a static fire).

4.2 Step 2 — Calculate

Press **Calculate**. The solver sends the form to `/calculate` and populates the results within a few seconds.



The results HUD (Performance Summary), the Warnings panel, and the design tables. In our example the Specific Impulse is 185.6 s, thrust 1000 N, chamber pressure 20 bar, mass flow 0.549 kg/s, and the minimum structural safety factor 2.89.

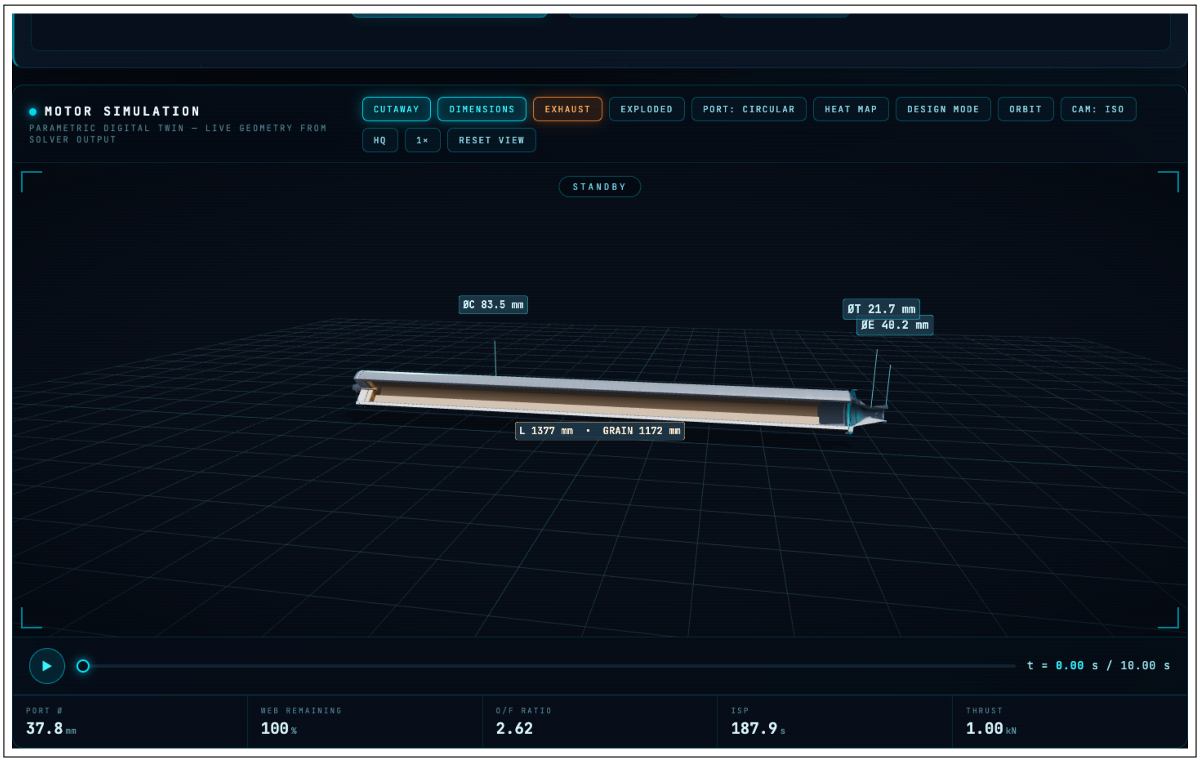
Reading the results:

- **Performance Summary (HUD):** The headline numbers — thrust, total impulse, Isp, c*, CF, chamber pressure, throat and exit diameters, mass flows, and a safety-state badge.
- **Warnings panel:** Design-criteria messages from the solver and the validation system. **Read these before trusting the numbers.** Our example raises three: c* just below its normal range (1325 m/s), a low expansion ratio ($\varepsilon = 3.4$), and a flash boiling risk. These are not errors; they tell you where the design sits at a limit and under which assumptions.
- **Design tables:** Full geometry — nozzle contour dimensions and angles, grain/port geometry, injector plan, wall thickness.

Warnings are a feature, not a defect. Rather than invent a number for something it cannot compute, HRMA warns explicitly or says "not modelled". When you see a red/orange warning, cross-check that quantity by hand or with an independent code.

4.3 Step 3 — Inspect the 3D model

Below the results is a 3D motor model built live from the geometry the solver produced. Use the cutaway, dimensions, exploded, heat-map and burn animation buttons to inspect it.



The 3D motor simulation (cutaway view). Dimensions come straight from the solver output: chamber diameter 83.5 mm, throat 21.7 mm, exit 40.2 mm, total length 1377 mm. The timeline at the bottom animates how the port diameter grows through the burn.

Interactive design mode. After a calculation, the chamber diameter, L* and expansion-ratio sliders recompute geometry in about a second; the 3D model and cross-section update live without a full recalculation.

4.4 Step 4 — Charts and cross-section

Performance charts (thrust curve, pressure, regression rate and port growth) are drawn in the app's dark theme, fully offline. The cross-section is a scaled engineering drawing generated from the same geometry the solver used.

The motor cross-section with a full legend: chamber wall, liner, fuel grain, injector plate, nozzle, and the final port diameter at burnout, all dimensioned in mm.

4.5 Step 5 — Export

The export buttons in the results panel produce: drawing and report PDFs, CAD files (including STEP cross-section mapping) and CSV/Excel data tables. Outputs are written to the `Documents/HRMA` folder.

5. Solid and Liquid Motors

The solid and liquid pages follow the same pattern; only the inputs are type-specific.

- **Solid motor (`/solid`):** Grain geometry (BATES, finocyl, slotted, etc.), number of segments, propellant family (KNDX, KNSB, APCP and more), ballistics and Monte Carlo uncertainty analysis. The burn-rate `a` and `n` coefficients come from a database.
- **Liquid motor (`/liquid`):** Propellant pair, feed system type and **power cycle** (pressure-fed, gas-generator, tap-off, staged combustion, FFSC, expander). A cycle power-balance solver checks whether the pump and turbine powers close. It also has supercritical CH_4/H_2 regenerative cooling and a gas-gas injector path. Combustion thermochemistry is solved with RocketCEA at the motor's *real* chamber pressure, O/F ratio and expansion ratio.

6. Scope and Limitations (Please Read)

HRMA is a capable preliminary-design and educational tool, but knowing its limits is essential:

- **Preliminary-design level.** The models are closed-form and 1D correlations. Turbopump maps, cavitation, rotordynamics, bearing life, startup/shutdown transients and 2D/3D conjugate heat transfer are *not* present.
- **Validation is limited.** Given the real operating point and the published geometry, liquid

vacuum Isp is reproduced with about $\sim 1.2\%$ median error over a 14-engine sample. That supports "predicts Isp well at a known point", *not* "predicts the flight performance of an unknown motor from scratch to that error". On the hybrid side the regression-rate and chamber-pressure errors are much larger (13–35%). For current figures see `docs/VALIDATION_STATUS.md`.

- **Not flight qualification.** HRMA alone does not support the final design, safety decision or qualification of a motor that will fly. That requires fatigue/creep analysis, physical testing and a formal V&V process.

The golden rule. HRMA is a starting point. Cross-check every critical result with an independent code (CEA / RPA / openMotor) and, where possible, with a static fire test. Knowing how to use the program is not, by itself, being a rocket engine engineer.

7. Automatic Updates

At startup HRMA checks GitHub Releases for a newer version. If one exists, a window appears: **Update now**, **Later**, or **Skip this version**. Choosing "Update now" downloads the installer, closes the app, installs silently, and reopens automatically.

New in version 2.6.25. This release brings three visible changes. (1) The *Cooling Channels* selection on the hybrid page now actually enters the calculation; in earlier versions the thermal analysis always ran uncooled regardless of what you picked. The warning shown when you select channels states that the coolant film coefficient is taken from literature and that coolant flow is not verified — that is a deliberate limit, not a fault. Chamber material and wall thickness now change the thermal results as well. (2) Warnings produced by the engines now appear as readable text in the interface language; previously some of them reached you as raw codes. (3) The update window shows release notes in your interface language.
Update note. On macOS the update install can take a few minutes, during which a notification is shown; do not open the app by hand until it finishes. If your network hits GitHub's request limit (possible on shared networks), the program now falls back to the ordinary release pages and still finds the update.

HRMA v2.6.25 — Rocket Motor Analysis Suite
github.com/berketez/HRMA